ACID-BASE CHEMISTRY: TITRAMETRIC DETERMINATION OF THE K_A OF A WEAK ACID, BUFFER EXPLORATION, AND ANALYSIS OF SALTS

OBJECTIVES

- To determine the K_a of a weak acid by recording its titration curve with a strong base.
- To explore the reaction of a buffer solution to the addition of acid and base.
- To study the effect of hydrolysis of salts on pH.

INTRODUCTION

One of the more important properties of an aqueous solution is its concentration of hydrogen ion. The H^+ (or H_3O^+) ion has great effect on the solubility of many inorganic and organic species, on the nature of complex metallic cations found in solutions, and on the rates of many chemical reactions. It is important that we know how to measure the concentration of hydrogen ion and understand its effect on solution properties.

For convenience, the concentration of H^+ ion is frequently expressed as the pH of the solution rather than as molarity. The pH of a solution is defined by the following equation:

$$pH = -log[H^+] \quad (1)$$

where the logarithm is taken to the base 10.

Basic solutions can also be described in terms of pH. In water solutions the following will always be obeyed:

$$[H^+] \times [OH^-] = K_w = 1.0 \times 10^{-14} \; at \; 25 \; ℃ \quad (2)$$

Since $[H^+]$ equals $[OH^-]$ in distilled (neutral) water, by Equation 2, $[H^+] = 1 \times 10^{-7} \; M$. Therefore, the pH of distilled water is 7. Solutions in which $[H^+] > [OH^-]$ are said to be acidic and will have a $pH < 7$; if $[H^+] < [OH^-]$, the solution is basic and its $pH > 7$. A solution with a pH of 10 will have $[H^+] = 1 \times 10^{-10} \; M$ and $[OH^-] = 1 \times 10^{-4} \; M$.

We may experimentally determine the pH of a solution in two ways. In the first of these we use a chemical called an indicator (usually a complex organic molecule), which is sensitive to pH. These substances have colors that change over a relatively short pH range (about 2 pH units) and can, when properly chosen, be used to roughly determine the pH of a solution. Two very common indicators are litmus, usually used on paper, and phenolphthalein, the most common indicator in acid-base titrations. Litmus changes from red to blue as the pH of a solution goes from about 6 to about 8. Phenolphthalein changes from colorless to red as the pH goes from 8 to 10. A given indicator is useful for determining pH only in the region in which it changes color. Indicators are available for

measurement of pH in all the important ranges of acidity and basicity. Universal indicators, which contain a mixture of several indicators and show color changes over a wide pH range, are also in common use.

Table 1: Common indicators, and the pH range in which they change color

INDICATOR	pH INTERVAL	COLOR CHANGE
thymol blue	1.2 - 2.8	red to yellow
orange IV	1.3 - 3.0	red to yellow
methyl orange	3.1 - 4.4	red-orange to yellow
congo red	3.0 - 5.0	blue to red
methyl red	4.6 - 6.2	red to yellow
litmus	4.5 - 8.3	red to blue
bromothymol blue	6.0 - 7.6	yellow to blue
thymol blue	8.0 - 9.6	yellow to blue
phenolphthalein	8.3 - 10.0	colorless to red
alizarin yellow R	10.0 - 12.0	yellow to red
indigo carmine	11.4 - 13.0	blue to yellow

The other method for finding pH is with a device called a pH meter. In this device an electrode sensitive to $[H^+]$ is immersed in a solution. This electrode generates a voltage related to the pH. The pH meter is designed so that the scale will directly furnish the pH of the solution.

Some acids and bases undergo substantial ionization in water and are called strong acids/bases because of their essentially complete ionization in reasonably dilute solutions. Other acids and bases, because of incomplete ionization (often only about 1% or less in 0.1 M solution), are called weak acids/bases. Hydrochloric acid, HCl, and sodium hydroxide, $NaOH$, are typical examples of a strong acid and a strong base, respectively. Acetic acid, CH_3COOH and ammonia, NH_3, sometimes delivered in the form of ammonium hydroxide (NH_4OH), are classic examples of a weak acid and a weak base.

A weak acid will ionize according to:

$$HA_{(aq)} \leftrightarrow H^+_{(aq)} + A^-_{(aq)} \tag{3}$$

At equilibrium,

$$\frac{[H^+][A^-]}{[HA]} = K_a \tag{4}$$

K_a is a constant characteristic of the acid HA; in solutions containing HA, the product of concentrations in the equation will remain constant at equilibrium independent of the manner in which the solution was made up. A similar relation can be written for solutions of a weak base.

Table 2: Weak Acids for Part A

Weak Acid	K_a
Potassium Bisulfate	1.2×10^{-2}
Monochloroacetic acid	1.4×10^{-3}
Formic acid	2.1×10^{-4}
Acetic acid	1.8×10^{-5}
Potassium hydrogen phthalate	3.9×10^{-6}

The value of the ionization constant K_a for a weak acid can be found experimentally in several ways. Perhaps the most obvious way is to simply measure the pH of a solution of the acid of known molarity. From the pH the $[H^+]$ can be calculated from (1). Equation (4) shows us that $[H^+]$ must be equal to $[A^-]$ since the $[H^+]$ ion is produced essentially from reaction (3) alone. The $[HA]$ in (4) is equal to the original molarity of HA, corrected if necessary for the amount of HA lost by the dissociation shown in reaction (3).

A very simple procedure that you will use in Part A involves very little calculation, is accurate, and does not even require a knowledge of the molarity of the acid. A sample of a weak acid HA, often a solid, is dissolved in water. The solution is titrated to generate a titration curve for the acid. During the titration, the HA is converted to A^- by the reaction:

$$OH^-_{(aq)} + HA_{(aq)} \rightarrow A^-_{(aq)} + H_2O_{(l)} \qquad (5)$$

At the equivalence point, the number of moles of A^- produced equals the number of moles HA originally in the untitrated sample. **Halfway to the equivalence point**, $[HA]$ equals $[A^-]$, so that by (4),

$$[H^+] = K_a \qquad (6)$$

By this method the $[H^+]$ obtained from the pH measurement is equal to the value of the ionization constant of the acid but only exactly halfway to the equivalence point.

Buffers

Some solutions, called buffers, are remarkably resistant to pH changes caused by the addition of an acid or base. These solutions contain both the salt of a weak acid or base and the parent acid or base ($[A^-]$ and $[HA]$ in our example). If a small amount of strong acid were added to that solution, the H^+will react with the A^- ion present, keeping the $[H^+]$ about where it was before the addition. Similarly, a small amount of strong base added to that solution will react with the HA present, producing some A^- ion and water but not appreciably changing the $[OH^-]$. If similar amounts of strong acid or base were added to pure water, the pH change could easily be several units. Commonly used antacids neutralize acid by acting as a buffer in your body.

To prepare a buffer that can maintain a given pH value, we need a solution of a weak acid or base and a solution of the salt of that acid or base. Given the value of K_a of the acid and the $[H^+]$ required in an acid buffer, we can substitute into (4) to find $\frac{[A^-]}{[HA]}$ in the buffer. By mixing appropriate volumes of the acid and salt solutions we can easily obtain that concentration ratio and prepare a buffer. A buffer will typically have a pH value whose associated H^+ concentration is near the K_a value for the weak acid in the buffer, since if the buffer is to be effective against addition of both acid and base, the ratio find $\frac{[A^-]}{[HA]}$ should be near one.

Hydrolysis

Hydrolysis means literally "water split." Salts which can be formed by the reaction of strong acids with strong bases, such as $NaCl$, KBr or $NaNO_3$, ionize completely when in aqueous solution, but the ions do not themselves react with the water. Thus, they form neutral solutions with a pH of about 7.

When salts of weak acids or bases are dissolved, however, they furnish ions which themselves react to some extent with the water (hydrolysis), regenerating small portions of the original weak acid or base and liberating some OH^- or H^+ in the process. These solutions then are not neutral, but characteristically acidic or basic.

For example, consider dissolving a few grains of sodium carbonate (Na_2CO_3) in water. As the grains dissolve, Na^+ and CO_3^{2-} ions dissipate into the solution. But some of the CO_3^{2-} ions, having a significant strength as a conjugate base, immediately react with water:

$$CO_{3\,(aq)}^{2-} + H_2O_{(l)} \leftrightarrow HCO_{3\,(aq)}^{-} + OH_{(aq)}^{-}$$

Since a significant amount of OH^- is produced, the sodium carbonate solution is noticeably basic ($pH > 7$). Salts containing the conjugate bases of weak acids produce basic solutions. For another example, the household cleaning agent TSP is trisodium phosphate (Na_3PO_4). When dissolved in water, the PO_4^{3-} hydrolyzes to produce HPO_4^{2-} along with considerable OH^-. This basic solution is very effective at attacking certain types of cleaning problems.

Hydrolysis causes the equivalence points of weak acid or weak base titrations to be distinctly basic or acidic. Only a strong acid titrated with a strong base (e.g., HCl and $NaOH$) will have an equivalence point near pH 7.

An important consequence of hydrolysis is the acidic pH of ordinary water which has been exposed to air. The trace of CO_2 in air readily dissolves in the water, reacting (hydrolyzing) to form carbonic acid

$$CO_{2\,(g)} + H_2O_{(l)} \leftrightarrow H_2CO_{3\,(aq)}$$

which in turn partially dissociates to liberate some H^+. For this reason, the pH of tap water is typically 5-6 and rainwater has a natural acidity (often enhanced, of course, by the additional acidity due to hydrolysis of oxide pollutants such as SO_2 and NO_2). Acidic rainwater readily dissolves many metal salts and leads to the transport of metals in soils and waterways. Also, for many of us, freshly boiled water tastes flat because it has not yet recovered its slight acidity for atmospheric CO_2. Bottled sparkling waters are enlivened by slight carbonation. Club soda represents the extreme case of sparkling fresh water.

PRE-LAB EXERCISES

1. Sketch a titration curve for a weak acid that is being titrated with a strong base. Indicate and label the initial pH, endpoint, equivalence point, buffer region and ½ the equivalence point on your curve.

2. Write out the Henderson-Hasselbach equation and indicate on which region of your sketched titration curve this equation would apply to calculating the pH.

You will also be testing the pH of three ionic compounds today: $NaCl$, NaF, NH_4Cl. You will need to prepare a 0.1 M solution of each.

3. Calculate how many grams are needed for each ionic compound to make 100 mL of a 0.1 M solution.

4. Write the balanced net ionic equation for each of the three ionic compounds. See Step 7 in Part C for an example. Examine the ions produced and predict if any of them will react with water to produce an acidic or basic solution.

PROCEDURE

NOTE: There are questions posed throughout the procedure sections of the manual which should be addressed in the conclusion section of your notebook.

A. Determining the K_a of a Weak Acid

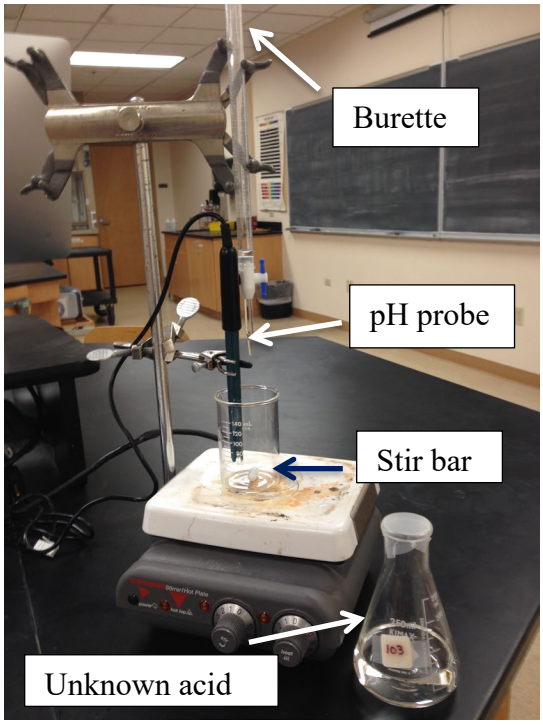


Figure 1: Titration setup.

1. Using the example in Figure 1, set up the titration apparatus including the pH probe.
2. Open Logger Pro on the computer with your pH probe plugged in. A pH reading should display in the upper left-hand corner.
3. Today we will be measuring changes in pH (y-axis) over time in minutes. Make appropriate adjustments under Experiment → Data Collection thinking also about an appropriate sampling rate. The collection time should be at least 10 minutes and 60 samples per minute.
4. Calibration: Use the buffer solutions provided to calibrate the apparatus by following step First, remove the pH probe from the storage solution bottle. Rinse the end of the probe with distilled water into a waste beaker.
5. Place the pH electrode into the pH 4 buffer solution. From the EXPERIMENT menu open CALIBRATE and select pH 1. Click the CALIBRATE NOW button. At the bottom of this CALIBRATION window find the KEEP button. Enter 4 (4 is the pH of buffer you are currently calibrating with) in the box above the KEEP button. When the voltage has stabilized, select KEEP.
6. Remove the electrode from the pH 4 buffer solution. Replace the cover on the pH 4 buffer solution. Rinse the electrode with distilled water. Pat the electrode dry with a Kimwipe.
7. Repeat for the pH 10 buffer solution. Enter 10 in the box above the second KEEP button. Rinse the electrode as before. When calibrations for both solutions have been done click on DONE.
8. Do not close the Logger Pro window or you will lose your calibration data.
9. Obtain from the cart a sample of a weak acid solution. **RECORD THE UNKNOWN NUMBER**. This solution is 0.1 M in one of the five weak acids listed in Table 2.

10. Be sure to follow all cleaning and rinsing skills you have learned previously. It is important to test your burette to ensure it is clean, and you can get a proper flow.
11. Measure out 20 mL of distilled water in a graduated cylinder and pour it into a clean 150 mL beaker. Place 20 mL of your acid sample in the water and add a clean stir bar Activate the magnetic stirrer.
12. Carefully clamp the pH electrode to the stand so that it is immersed into your acid/water stirring solution. Add 2-3 drops of phenolphthalein or any other appropriate indicator to this solution to visually detect the anticipated end point of the titration.
13. Making sure the stopcock is closed, use a funnel to slowly fill the burette of the apparatus with $0.200\ M\ NaOH$ slightly above the 0 mL mark. Set the burette high enough to fit a 50 mL beaker between it and the 150 mL beaker. While catching the $NaOH$ in the 50 mL beaker, adjust the delivery rate of the burette to approximately 1 drop/sec.
14. When the burette level approaches the 0 mL mark move the 50 mL beaker so the $NaOH$ can be delivered to the stirring acid solution while simultaneously starting the data collection by selecting COLLECT.
15. Titrate the solution to well beyond the equivalence point and stop the data collection by selecting STOP when appropriate (the collection should complete in ~5 to 10 minutes). Pay attention to the flow rate from your burette. If it slows down noticeably during the titration, let your instructor know.
16. When you start your next data collection, after you hit collect, select Store Latest Run to save this titration curve. The next curve you generate can be added on the same plot. These titration curves should be included in your data section of your submitted lab notebook.
17. Under the ANALYZE menu, select the Examine option. Using the cursor, determine the pH of **halfway to the equivalence point** in the titration and **determine the K_a** of the acid. (Hint: You can use time to the equivalence point and divide by 2 to estimate the halfway to the equivalence point.) **Of the possible acids listed in Table 2, identify which acid you think you were given**?
18. Repeat the titration for a second trial. Compare your results to those of the first trial. Before discarding this solution, you may want to explore what happens to the pH when you add a small amount of HCl solution. While collecting pH data, add 6 M HCl solution with a dropper until the pH drops dramatically. Can you pass through the equivalence point several times by adding HCl and $NaOH$ solutions?

B. Creating a Buffer

1. Repeat the Part A titration procedure. However, when the pH of the titration reaches that of the **halfway to the equivalence point** (measured above), close the stopcock of the $NaOH$ burette. Here, $\frac{[A^-]}{[HA]}$ should be equal to one, refer to your titration curve you drew previously.

2. While still recording the pH of the solution on the computer, (you may need to restart the data collection procedure) add 5 drops of 1M $NaOH$ to your buffer solution and stir thoroughly. Note the pH of the mixed solution. Now, add 5 drops of 1M HCl and stir thoroughly. Measure the pH of this solution. Finally, add another 5 drops of 1M HCl and note the new pH.
3. Rinse your pH probe with distilled water and then pat dry with a Kimwipe. Place the pH probe in the pH 7 buffer solution for about 15 seconds. Replace the cover on the pH 7 buffer solution. Rinse the probe with distilled water and pat dry with a Kimwipe.
4. Measure the pH of the distilled water in the laboratory. Add 50 mL of distilled water to a 150 mL beaker. To the sample of water (volume similar to that of the titrations) add 5 drops 1M $NaOH$, then 5 drops 1M HCl and then 5 more drops of the 1M HCl. Between each addition and after allowing the solution to mix, measure and record the pH of the solution.
5. What can you conclude about the capacity of your buffer to absorb acid or base? Why isn't the pH of distilled water exactly 7?
6. Rewrap your unknown and return it to the cart.

C. Hydrolysis of Salts

1. In this portion of the lab three salts will be hydrolyzed. The first will be sodium chloride ($NaCl$), the second will be ammonium chloride (NH_4Cl), and the third will be sodium fluoride (NaF). You should observe one salt doesn't significantly change the pH, one will LOWER the pH and one will RAISE the pH.
2. Weigh out the calculated amounts of the three salts (see Pre-Lab calculations), making sure to label the samples. REMEMBER to cap bottles tightly, use clean scoopulas, throw away used weigh boats, wash used scoopulas, and clean up any spills.
3. Fill three 150 mL beakers with 100 mL of Ionic Strength Adjuster measured using a graduated cylinder. Label each for the salt you will put in it. NOTE: DO NOT ADD THE SALT YET.
4. Rinse your pH probe with distilled water and pat dry with a Kimwipe. Place the pH probe in the pH 7 buffer solution for about 15 seconds. Rinse the probe with distilled water and pat dry with a Kimwipe.
5. Place the pH probe in the first 150 mL beaker and select COLLECT in Logger Pro.
6. Slowly pour the first salt into the beaker while monitoring the pH of the solution with the computer.
7. Determine the final pH of the 0.1 M solution of the salt. **Write a set of three reactions to describe the acid-base chemistry** between the salt and water to rationalize the observed pH.

Here is an example set of three reactions for dissolving $MgBr_2$:

a. $MgBr_2(s) \xrightarrow{H_2O} Mg^{2+}(aq) + 2\,Br^-(aq)$

   i. $Mg^{2+}(aq) + 2\,H_2O(l) \leftrightarrow Mg(OH)_2(aq) + 2H^+(aq)$

   ii. $Br^-(aq) + H_2O(l) \leftrightarrow HBr(aq) + OH^-(aq)$

In this example, the two products in i. are magnesium hydroxide (a weak base) and H^+ (a strong acid). The strong acid will overpower the weak base and magnesium ions make the solution acidic. In ii., the products are hydrobromic acid (a strong acid) and hydroxide ions (a strong base), so the bromine ions do not alter the pH and are therefore, considered neutral ions. Overall, this salt (Mg^{2+} and Br^-) should cause the solution to be ACIDIC.

8. Repeat this process for the remaining two salts.

WASTE AND CLEANUP

a. THOROUGHLY rinse your burette with distilled water.

b. Return the pH probe to the plastic storage solution bottle. Be sure the top is on firmly and the solution in the bottle will cover the probe. If there is not enough solution see you instructor or TA for more.

c. Return the pH probe to the cart in its box in an upright position (with the pH probe immersed in the storage solution). Rewrap the Go!Link, with a rubber band. Return it and the Go!Link adapter to the appropriate tub on the supply cart.

d. Put all waste solutions in the waste container in the hood and clean the remaining glassware with soap and an appropriate brush or scrubby.

CONCLUSION

1. Include a discussion of your titration curves. Report the $\boldsymbol{K_a}$ and identify your unknown acid.
2. Report the observed changes in pH of your buffer and distilled water when acid or base was added. Did these changes occur as expected?
3. Be sure to include a discussion on how dissolving salts can cause changes to the measured pH.